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Perceptual Threshold of Round-Trip Delay in TPTA Systems during Viscoelastic Object Interaction

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Overview

- 1 Motivation
- 2 Objective
- 3 Methodology
- 4 Results
- 5 Conclusion and Future Work
- 6 References

Motivation

- In a telepresence and teleaction (TPTA) system, a human operator explores and interacts with a remote environment via a haptic interface/device.
- In that, the motion and force information is transmitted over a communication channel that introduces stochastic **time delays** between the leader and follower.
- **Forward delay**: delays motion commands from the leader to the follower.
- **Backward delay**: delays force feedback from the follower to the leader.

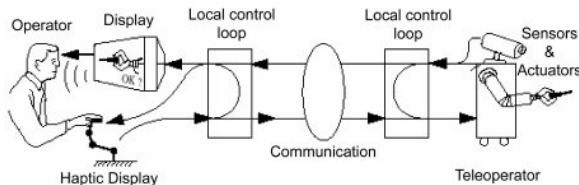


Fig. 1. General overview of a TPTA system ¹

¹Hinterseer et al., 2008

Objective

- **Round-trip delay** ($T = T_f + T_b$) influences the operator's perception of the remote environment.
- For instance,
 - Delayed force feedback underestimates perceived stiffness, damping, and inertia (mass). ²
 - Perceived stiffness decreases with increasing delay in force. ^{3 4}
 - Opposite trend: perceived stiffness increases with delay. ⁵
 - Delay affects damping but not stiffness perception. ⁶
- Determine the **permissible variation in round-trip delay** that remains imperceptible to the operator.
- **In this study:** We estimate the **Just Noticeable Difference (JND)** for round-trip delay in a TPTA system during viscoelastic objects interaction.

²Fu et al., 2018

³Knörlein et al. 2009

⁴Ohnishi and Mochizuki, 2007

⁵Pressman et al., 2007

⁶Colonnese et al., 2015

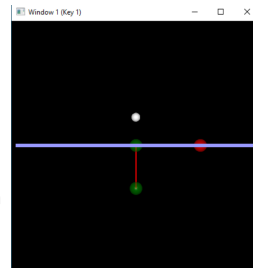
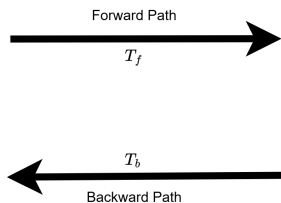
Methodology

A. Setup

- For setup, we utilized a typical TPTA/teleoperation scenario in a virtual environment. ⁷
- **Operator:** 'PHANToM Premium 1.5 HF' haptic device → Virtual tool (white ball)
- **Teleoperator:** Virtual Environment (Viscoelastic plane $k = 200$ N/m, $b = 1$ N-s/m)



Operator



Teleoperator

Fig. 2. Experimental setup of the virtual TPTA system

⁷A. Bhardwaj et al., 2017

Methodology

- **Delay implementation:** A queue-based buffer introduces the round-trip delay.
- **Round-trip delay:** $T = T_f + T_b$, with T_f in the forward path and T_b in the backward path.
- **Force rendering:** Interaction force $F_z(t)$ is computed using a Kelvin-Voigt model.

$$F_z(t) = \begin{cases} 0 & \text{if } z_{tool}(t - T_f) \geq z_{plane} \\ -k(z_{tool}(t - T_f) - z_{plane}) - b\dot{z}_{tool}(t - T_f) & \text{if } z_{tool}(t - T_f) < z_{plane} \end{cases} \quad (1)$$

z_{tool} = position of the virtual tool along the z-axis

z_{plane} = position of the virtual plane along the z-axis.

- **Delayed force feedback:** F_z is computed from delayed position and delayed again by T_b .
- **Added latency:** The total delay T represents the latency between motion and force feedback.

B. Design and Procedure

- **Active exploration:** Users explore the virtual plane using a haptic stylus.
- **Controlled motion:** Users follow a sinusoidal trajectory at **1 Hz**.
- **Visual guidance:** A red reference ball appears upon contact with the virtual plane.
- **Psychophysical paradigm:** Transformed adaptive **3I-3AFC** method.⁸
- **Trial structure:** Three sequential interaction windows (W_1 , W_2 , W_3).
- **Delay assignment:** Two windows contain reference delay T_r ; One randomly contains a comparison delay T_c .
- **Task:** Identify the **odd-one-out** using '1', '2', or '3'.

Trial Structure

$$W_1 \longrightarrow W_2 \longrightarrow W_3$$

Two intervals: T_r One interval: T_c

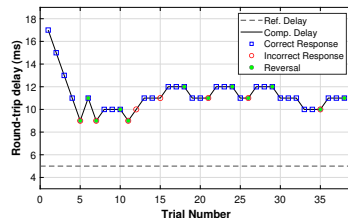


Fig. 3. Example trial of the transformed adaptive staircase.

⁸L. A. Jones et al. 2012

Methodology

- **Adaptive T_c :** Adjusted across trials to converge on the discrimination threshold.
- **Stage 1:** 1-up-1-down; step size = 2 ms; until the **2nd reversal**.
- **Reversal:** Change in the direction of T_c adjustment.
- **Stage 2:** 1-up-3-down; step size = 1 ms; until **12 reversals**.
- **Convergence level:** 1-up-3-down $\rightarrow$ **79.4% correct**.

C. Data Collection

- **Participants:** 10 users.
- **Ethics:** Approved by the Institutional Ethics Committee, IIT Jodhpur.
- **Three reference delays conditions:** $T_r = 0, 5$, and 10 ms, presented in randomized order.
- **Duration:** Approximately 1.5-2 hour per participant.

Results

JND Estimation:

JND = mean ($T_c - T_r$) over the **last 6 reversals**.

JND across users:

- $T_{r1} = 0$ ms: [4.33, 14.33] ms $\rightarrow 8.60 \pm 2.94$ ms
- $T_{r2} = 5$ ms: [3.33, 11.17] ms $\rightarrow 6.92 \pm 2.37$ ms
- $T_{r3} = 10$ ms: [2.67, 8.67] ms $\rightarrow 6.02 \pm 1.90$ ms

Observation: JND decreases as T_r increases.

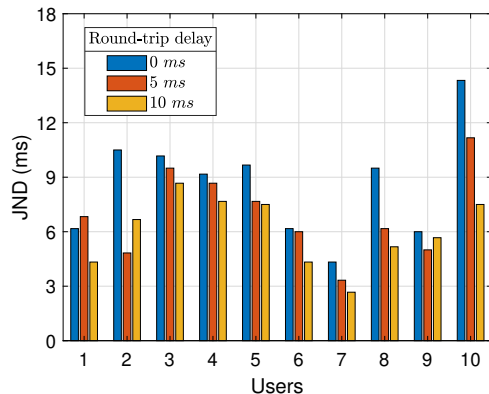


Fig. 4. JND across users for three reference delays.

Results

Statistical Analysis:

- **One-way repeated-measures ANOVA**

- Significant main effect of reference delay

$$F(2, 18) = 9.911, \quad p = 0.008, \quad \eta_p^2 = 0.546$$

- **Post-hoc: Bonferroni-corrected pairwise comparisons**

$$JND_{0 \text{ ms}} > JND_{10 \text{ ms}}, \quad p = 0.005$$

- No significant differences:

$$JND_{0 \text{ ms}} \neq JND_{5 \text{ ms}}, \quad JND_{5 \text{ ms}} \neq JND_{10 \text{ ms}}.$$

Key finding: Reference delay significantly affects JND.

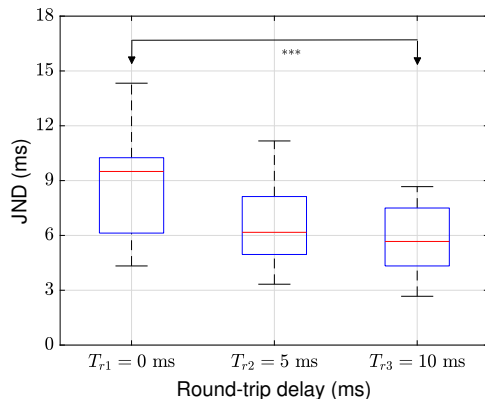


Fig. 5. Distribution of JND across reference delays.

Conclusion and Future Work

Conclusion

- **Reference delay affects JND:** JND significantly decreases at $T_r = 10$ ms compared with $T_r = 0$ ms.
- **Increased sensitivity:** Higher reference delays enable detection of smaller delay differences; $T_r = 5$ ms showed no significant difference.
- **Possible explanation:** Greater baseline latency makes additional delay variations more perceptually salient.

Future Work

- **Higher delays:** Introduce stabilization techniques to investigate JND at higher reference delays.
- **Participant diversity:** Increase gender and age diversity and account for inter-subject variability.
- **Movement conditions:** Investigate the effects of different movement frequencies (0.5 and 2 Hz).
- **Asymmetric delays:** Examine the effects of independently varying forward (T_f) and backward (T_b) delays.

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Thank You!



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